

Introduction

The Arctic food web is a complex network of predator-prey relationships that sustains life in one of Earth's most extreme environments. While ecologists often study these interactions through observation and field data, mathematical models can help provide a deeper understanding of how energy flows through the system. As someone who has studied environmental science, ecosystem studies have always been an interest of mine. Using concepts we learned in linear algebra and the articles provided, I aim to analyze the Arctic food web by modeling predator-prey relationships and identifying key species. This approach will help us better understand the Arctic food web's structure and its vulnerability to environmental changes.

Data Collection

For my Arctic food web analysis, I utilized data from earthreminder.com. The ecosystem includes a variety of species, such as baleen whales, plankton, Arctic cod, polar bears, krill, walruses, orcas, penguins, leopard seals, and silverfish. These species were selected based on their ecological roles within the food web, representing a range of trophic levels (the hierarchical levels in an ecosystem, representing different positions in the food chain based on their feeding relationships), from primary producers to apex predators.

Data

	Preys on	Serves as prey for
Plankton	N/A	Baleen Whales, Silverfish, Arctic cod
Krill	N/A	Baleen Whales, Silverfish, Arctic cod
Arctic Cod	Krill, Plankton	Leopard Seals, Walruses, Penguins
Silverfish	Krill, Plankton	Leopard Seals, Walruses, Penguins
Baleen Whale	Plankton, Krill	Orcas
Walrus	Arctic Cod, Silverfish	Orcas
Penguin	Arctic Cod, Silverfish	Orcas, Polar Bears
Leopard Seal	Arctic Cod, Silverfish	Orcas, Polar Bears
Orca	Penguins, Walruses, Leopard Seals, Baleen Whales	N/A
Polar Bear	Leopard Seal, Penguins	N/A

Table 1. Feeding Relationships of Arctic Species

Data Analysis

Using the data we collected in Table 1, we can create a diagram that illustrates all the feeding relationships and thus the food web as a whole.

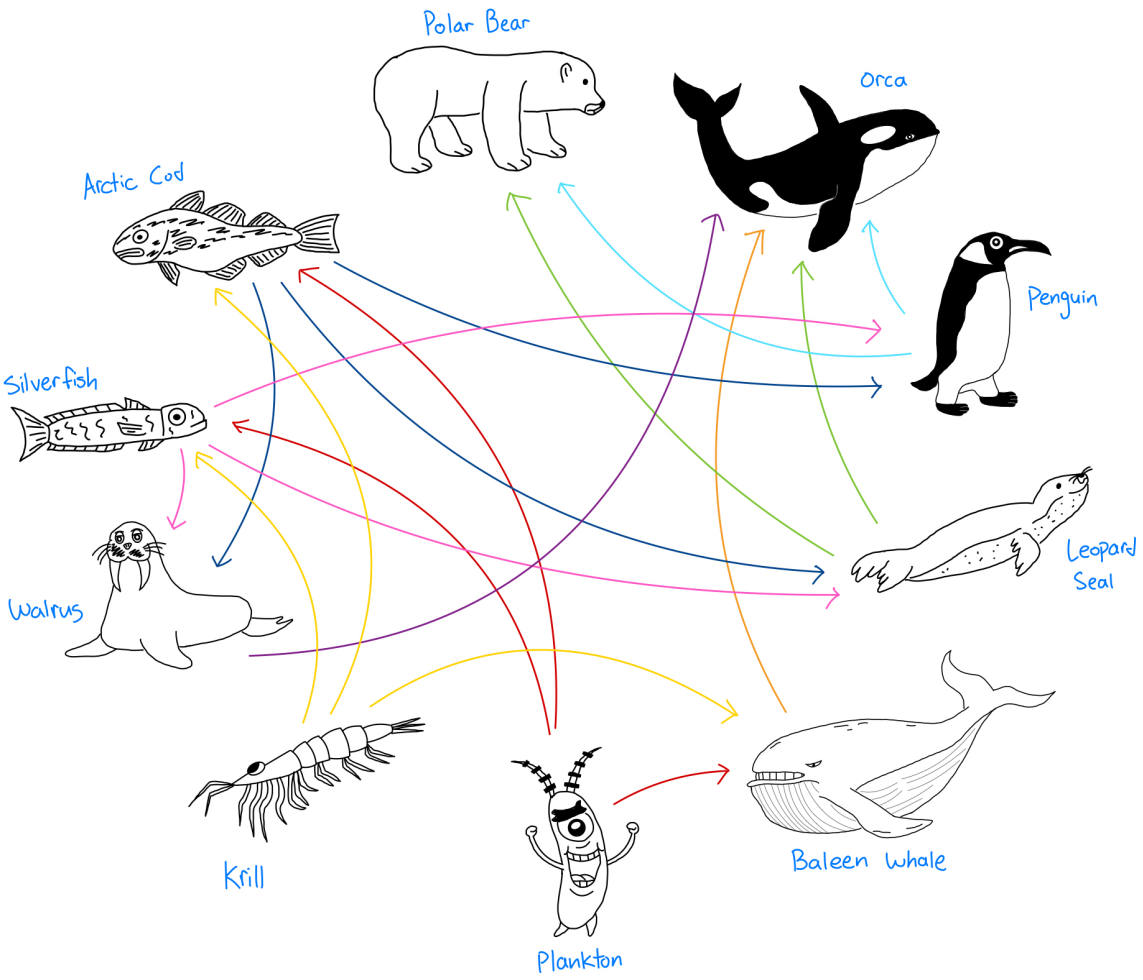


Figure 1. Diagram of Food Web

With the data collected from Table 1 and Figure 1, we can now create the adjacency matrix M. First, we can set it up using a table.

	Plankton	Krill	Arctic Cod	Silver fish	Baleen Whale	Walrus	Penguin	Leopard Seal	Orca	Polar Bear
Plankton	0	0	0	0	0	0	0	0	0	0
Krill	0	0	0	0	0	0	0	0	0	0
Arctic Cod	1	1	0	0	0	0	0	0	0	0
Silverfish	1	1	0	0	0	0	0	0	0	0
Baleen Whale	1	1	0	0	0	0	0	0	0	0
Walrus	0	0	1	1	0	0	0	0	0	0
Penguin	0	0	1	1	0	0	0	0	0	0

Leopard Seal	0	0	1	1	0	0	0	0	0	0
Orca	0	0	0	0	1	1	1	1	0	0
Polar Bear	0	0	0	0	0	0	1	1	0	0

Table 2. Adjacency Matrix Setup

Now, using Table 2, we can create the adjacency matrix M.

$$M = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

Figure 2. Matrix M

	Sum of values in row	Sum of values in column
Plankton	0	3
Krill	0	3
Arctic Cod	2	3
Silverfish	2	3
Baleen Whale	2	1
Walrus	2	1

Penguin	2	2
Leopard Seal	2	2
Orca	4	0
Polar Bear	2	0

Table 3. Row and Column Sums of Matrix M

The adjacency matrix M represents the direct predator-prey relationships in the Arctic food web. Each row corresponds to a predator, and each column corresponds to prey. A value of 1 indicates that the predator eats the prey, while 0 means there is no direct predation.

The row sum for each species indicates how many other species it directly dominates (eats). Plankton and Krill have row sums of 0, which makes sense because they are at the base of the food web and do not prey on any other species. On the other hand, Orca has the highest row sum of 4, meaning it directly dominates the most species in the food web. This makes the Orca a top or apex predator in the Arctic ecosystem.

The column sum for each species represents the number of species that directly dominate a given species. In other words, it shows how many predators a species has in the food web. For example, the column sum for plankton is 3, meaning that it is directly preyed on by 3 species. Meanwhile, orca and polar bear have column sums of 0 meaning that the species have no predators.

Next is the matrix M^2, which we can calculate using Figure 2.

$$M^2 = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 3 & 3 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Figure 3. Matrix M^2

	Sum of values in row	Sum of values in column
Plankton	0	7
Krill	0	7
Arctic Cod	0	5
Silverfish	0	5
Baleen Whale	0	0
Walrus	4	0
Penguin	4	0
Leopard Seal	4	0
Orca	8	0
Polar Bear	4	0

Table 4. Row and Column Sums of Matrix M^2

The matrix M^2 represents indirect dominance in the Arctic food web. Each entry indicates the number of 2-step paths from Species A to Species B, meaning Species A indirectly dominates Species B through an intermediate species. For example, orca indirectly

dominates plankton through baleen whales because orcas prey on baleen whales, who prey on plankton. By analyzing M^2, we can dig deeper into the structure of the Arctic food web, revealing how species influence each other beyond just direct predation. This helps identify hidden keystone species (organisms that other species in the ecosystem depend upon and whose disappearance would devastate their biological communities) and get a better understanding of the ecosystem's stability.

If we calculate the row sums once again, we notice that Orca has the highest indirect dominance of 8, influencing 8 species through intermediate steps. Walrus, penguin, leopard seal, and polar bear each have an indirect dominance of 4, showing their significant role in the food web. The species with a row sum of 0 have no indirect dominance.

The column sums of M^2 represent the number of species that indirectly dominate a given species through intermediate steps. This shows how many predators influence a species through indirect pathways. The column sum for plankton is 7, meaning that it is indirectly dominated by 7 species.

Finally, the matrix $A = M + M^2$, which we can calculate by adding Figure 2 and 3.

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 2 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 2 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 2 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 3 & 3 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 2 & 2 & 0 & 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

Figure 4. Matrix $A = M + M^2$

	Sum of values in row	Sum of values in column
Plankton	0	10
Krill	0	10

Arctic Cod	2	8
Silverfish	2	8
Baleen Whale	2	1
Walrus	6	1
Penguin	6	2
Leopard Seal	6	2
Orca	12	0
Polar Bear	6	0

Table 5. Row and Column Sums of Matrix $A = M + M^2$

The matrix $A = M + M^2$ represents total dominance in the Arctic food web, combining both direct and indirect dominance. Each entry indicates the total number of ways Species A dominates Species B, either directly (through predation) or indirectly (through intermediate species).

By calculating the row sums, we notice that orca has the highest total dominance of 12, meaning it directly and indirectly dominates the most species in the food web. Since they are at the base, plankton and krill have no dominance within the food web, evidenced by their row sums of 0.

The column sums of $M + M^2$ represent the total number of species that dominate a given species, both directly and indirectly. This provides a comprehensive view of how many species influence a species through all possible pathways. The column sum of krill is 10, meaning it is dominated both directly and indirectly by 10 species in total.

Conclusion

The analysis of the Arctic food web using linear algebra and graph theory reveals several key insights into the structure and dynamics of the ecosystem. The orca emerges as the most dominant species, with a total dominance score of 12 (by far the highest out of any species), indicating it directly and indirectly influences the most species in the food web. This makes the orca a keystone species, as its presence or absence would significantly impact the ecosystem as a whole. The polar bear is the only other species with a column sum of 0 on the total dominance matrix, while having a total dominance score of 6. Similar to the orca, this reflects its influence as an apex predator.

Species like the Arctic cod, while not a top predator, is a crucial middle-tier species, serving as a primary food source for many predators, including walruses, penguins, and leopard seals. If the Arctic cod were removed from the food web, it would disrupt the energy flow, potentially leading to a series of effects on both predators and prey. This highlights the importance of indirect relationships, such as orca→baleen whales→plankton,

which contribute to the stability of the ecosystem by creating multiple pathways for energy transfer.

The basal species, plankton and krill, have no dominance but are heavily preyed upon, with column sums of 10 in the total dominance matrix. This underscores their role as the foundation of the food web, supporting all higher trophic levels. The walrus, penguin, and leopard seal each have a total dominance score of 6, while also having column sums of 1, 2, and 2. This reflects their dual roles as both predators and prey, which helps maintain the balance of the ecosystem.

Limitations & Possible Extensions

Although this analysis through the scope of linear algebra provides a deeper understanding of the Arctic food web, it inevitably has limitations.

- The model includes only a subset of species and interactions, omitting many organisms (bacteria, algae, etc.) and complex relationships that exist in the real world Arctic ecosystem.
- The model assumes that predator-prey relationships are constant, but real ecosystems are dynamic and can change due to environmental factors, migration, or population fluctuations.
- The model treats all predator-prey relationships as equal, ignoring the varying strengths of these interactions (Species A could prey on Species B a lot more than it does on Species C).

Using these limitations, we can brainstorm some possible extensions for this project such as including more species, incorporating weighted relationships, and dynamic modeling. Additionally, I would love to analyze the impacts of climate change, investigating how rising temperatures, melting ice, and other climate change effects could alter the structure and stability of the Arctic food web.